

Read the text below. For questions (6—10) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

A WARMING LIFE JACKET

New liner contains a substance that helps fight heat loss in chilly water.

Sometimes the biggest threat from a boat sinking isn't the accident itself. It's not even the sharks that might be swimming nearby. It's a life-threatening loss of body heat from remaining too long in cold water. Now, a South African teen has invented a heat-producing liner for life jackets. It could help delay injuries — or death — until a rescue is possible.

Normal body temperature for people is around 37° Celsius (98.6° Fahrenheit). But when the core body temperature falls below 35°C (95° F), people suffer from something called hypothermia. When this occurs, the body doesn't function quite the way it should, says Danielle Mallabone. She is a 17-year-old junior at St Teresa's High School in Johannesburg, South Africa.

With mild hypothermia, blood vessels just beneath the skin shrink. This restricts blood flow to help cut the loss of heat from blood. (As blood cools, it speeds the cooling of internal tissues.) Hypothermia also triggers shivering. Those muscle contractions help generate heat to somewhat boost the body's internal temperature, she notes.

During severe hypothermia, things get much worse. People become confused and uncoordinated. They also have difficulty speaking. Eventually, major organ systems such as the heart will fail. This can lead to death.

The body's temperature can drop to dangerous levels even in relatively warm water, explains Mallabone. That's why she designed a heat-producing liner for life jackets. Pockets in the liner hold a powdered chemical called calcium oxide, which gives off heat when it gets wet.

That heat-producing, or exothermic, reaction warms the water between the life jacket and someone's body. This might stave off severe hypothermia long enough for a rescue.

Mallabone tested her own invention by jumping into 10°C (50°F) water. Each test lasted an hour. The first version of her life jacket liner included only 50 grams of calcium oxide, she notes. «But that amount didn't provide enough heat, and my body temperature dropped to 35°C after just an hour», she says. So, the next version included 1 kilogram (2.2 pounds) of the heat-producing chemical. In her test using that liner, her body temperature stayed above 36.1°C.

The chemical reaction between calcium oxide and water generates heat slowly. The reaction began producing heat after 5 minutes, Mallabone found. The liner produced the most heat about 25 minutes after the life jacket was first immersed.

The teen presented her findings on May 13 in Phoenix, Ariz., at the Intel International Science and Engineering Fair. The Society for Science & the Public, which created the fair in 1950, still runs the competition. (SSP also publishes Science News for Kids.)

Overall, Mallabone's tests suggest there's no risk the jacket liner will explode or heat up so much that it risks causing burns. The reaction also doesn't produce acidic byproducts. And because calcium oxide doesn't react with humidity (water vapour in the air), the liners can be stored for long periods and still work when needed.

- 6 What is the most dangerous thing when you have a boat accident?
 - A The wreckage of the vessel.
 - B The sharks swimming in the water.
 - C The loss of body heat.
 - D The cold water.
- 7 What happens during hypothermia?
 - A People become more concentrated.
 - B People's movements become uncoordinated.
 - C The body's temperature rises.
 - D The blood flow stops.
- 8 What does a new liner contain?
 - A A substance that helps fight high temperature.
 - B Calcium dioxide which gives off heat.
 - C Pockets with baking powder.
 - D A solid substance to stop cooling.
- 9 How long did it take to test the invention?
 - A A month.
 - B Several hours.
 - C A week.
 - D Five minutes.
- 10 What are the benefits of the warming life jacket?
 - A It can create a heat-producing chemical.
 - B It can cause a hypothermic reaction.
 - C It can be stocked continually.
 - D It functions with humidity.

Task 3

Read the text below. Match choices (A—H) to (11—16). There are two choices you do not need to use. Write your answers on the separate answer sheet.

SPECTACULAR TRIPS THAT COUPLES SHOULD TAKE

- 11 **French Polynesia**
You want to relax, he wants to stay active — but you both crave for pristine sand and azure waters. This island nation can please all. Nelson suggests spending three nights unwinding and chilling out in Bora Bora, then taking in four nights in Moorea, which offers a little more action, like tours, hiking and some nightlife. French Polynesia has some of the most crystal blue water in the world. Couple that with over-the-water bungalows, and you and your honey are in for beautiful views all day long.
- 12 **Australia**
If you believe variety is the spice of life, Down Under is where you'd want to go for some zing. World-renowned wineries and extensive arts offerings are perfect for artsy couples, while history buffs will enjoy the Penal Colony town of Port Arthur, Tasmania, along with aboriginal experiences in Queensland, Nelson says. Nature enthusiasts can pass their time at the Great Barrier Reef. Spend your trip in Port Douglas, Sydney and Hobart. Australia is a great destination for couples because it has everything.
- 13 **Puerto Vallarta**
Considered one of the best honeymoon locations in the world, this charming city in Mexico features magnificent scenery, a cobblestone-lined city centre, a sophisticated arts scene and a number of restaurants, nightclubs and lounges. Enjoy a day napping in the sunshine and the next few days golfing, jet skiing or going deep-sea fishing. The town is compact, so almost everything you'd want to do and to see is within walking distance or an inexpensive taxi ride away.
- 14 **Montana Fly Fishing**
If you crave a unique outdoor experience, give fly-fishing a try. The Triple Creek Ranch in Darby, Montana, caters to all levels — from a pond where guides teach beginners, to a float down the Bitterroot River where experts can join the guides who engineered the mechanical fish in *A River Runs Through It*, says Betsy Donley, a travel specialist with Camelback Odyssey Travel in Phoenix. «I've planned fly fishing trips with couples. It's always fun, as there is lots of camaraderie and light-heartedness».
- 15 **Paris**
It may sound cliché, but the City of Light lives up to its romantic reputation that goes beyond the harmonic language and champagne. Enjoy a scenic picnic at Sacre Coeur, take a ride down the Seine River, stand in awe of the Notre Dame and stroll the eclectic Latin Quarter, with its bookstores and funky restaurants fueled by the energy from the Sorbonne. See the Eiffel Tower from another perspective with a cozy dinner at Jules Verne, the luxurious restaurant on the tower's second floor by Michelin starred chef Alain Ducasse.
- 16 **Cape Town, South Africa**
Boasting a stellar wine country and beautiful beaches, this city is known for its vino, culinary prowess and the friendliness of its people. Geographically, it's like Napa Valley and Santa Monica somehow magically merged. From Cape Town, head to the bush for a few days on safari. The Sabi Sabi Private Game Reserve is stunning and the best way to see the wild.

You will go there if you _____

- A are interested in observing animals in the wild.
- B dream of indulging into a romantic atmosphere of lights, river rides and restaurants.
- C have just got married and don't want to spend much money.
- D and your partner are looking for variety.

- E want to slim in the purest water.
- F desire to have a new unexpected undertaking.
- G enjoy calmness and silence.
- H prefer spending time in a comfortable hotel.

Task 4

Read the text below. Choose from (A—H) the one that best fits each space (17—22). There are two choices you do not need to use. Write your answers on the separate answer sheet.

Walking barefoot, (17) _____, has gone from being a crazy counter-culture trend to a scientifically-researched practice with a number of remarkable health advantages, such as increasing antioxidants, reducing inflammation, and improving sleep.

Earthing means walking barefoot on soil, grass or sand (i.e. any natural surface). So, we'll have to get off the sidewalk. Early studies are showing that the health benefits come from the relationship (18) _____. The planet has its own natural charge, and we seem to do better (19) _____.

A review published in *the Journal of Environmental and Public Health* looked at a number of studies (20) _____. In one, chronic pain patients using grounded carbon fiber mattresses slept better and experienced less pain.

Another study found that earthing changed the electrical activity in the brain, (21) _____. Still other research found that grounding benefitted skin conductivity, moderated heart rate variability, improved glucose regulation, reduced stress and boosted immunity.

One particularly compelling investigation, published in *The Journal of Alternative and Complementary Medicine*, found that earthing increases the surface charge of red blood cells. As a result, the cells avoid clumping, (22) _____. Another study in the same journal found that earthing may help regulate both the endocrine and nervous systems.

- A when we're in direct contact with it
- B also known as «earthing»
- C as measured by electroencephalograms
- D that highlight how drawing electrons from the earth improves health
- E Even if there were no proven benefits
- F which decreases blood viscosity
- G between our bodies and the electrons in the earth
- H can reduce cancer risk

USE OF ENGLISH

Task 5

Read the text below. For questions (23—32) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

If you're the type to forget (23) _____ dates but your other half isn't, it might be wise to (24) _____ in this piece of relationship-saving jewellery.

The Remember Ring is a nifty little gadget that (25) _____ a normal ring from the outside, but inside uses Hot Spot technology to heat itself up a handy 24 hours (26) _____ your anniversary.

Every hour the (27) _____ of metal gets hotter and hotter for 10 seconds, continuing to warm until, at 120°F, it becomes impossible to ignore, reminding you that a special day is around the (28) _____.

Fitted with a battery-charged micro-chip clock that converts heat in your hand into electricity, manufacturers promise the ring is maintenance free and will never (29) _____.

«Using a micro thermopile, The Remember Ring converts the heat from your hand into electricity, (30) _____ the battery charged and microchip clock running perpetually», say makers *Alaska Jewelry*. «Just specify your anniversary date when you order, and we'll program your ring for you. (31) _____ it and forget it — until your anniversary!»

The ring costs \$760 (£497), has a lifelong (32) _____, and comes in seven styles, available in 14 carat white and yellow gold in men's sizes 10 to 13.

	A	B	C	D
23	important	large	trivial	relevant
24	advance	invest	plunge	divest
25	doubles	differs	relates	resembles
26	after	until	at	before
27	lump	spot	piece	part
28	door	corner	floor	block
29	put off	take off	switch off	be off
30	holding	keeping	managing	storing
31	Set	Apply	Settle	Stick
32	security	contract	warranty	break

Task 6

Read the texts below. For questions (33—42) choose the correct answer (A, B, C or D). Write your answers on the separate answer sheet.

Last year, over 200,000 students (33) _____ SAT Program tests in over 175 countries outside the US. The SAT and SAT Subject Tests (34) _____ overseas six times a year: in October, November, December, January, May, and June.

International test takers can choose from more than 1,000 international test centres (35) _____ online in the Student area of the website. A list of international test centres (36) _____ in the international edition of the *The SAT Codelist, International Edition*. Occasionally, supplementary test centres are opened on request where and when necessary. In most countries, students (37) _____ more than 121 km (75 ml) from the nearest test centre can request that we open a test centre (38) _____ to their home.

	A	B	C	D
33	taken	took	take	has taken
34	offered	were offered	are offered	had been offered
35	listed	list	to list	listing
36	are also provided	will provide	also provided	is also provided
37	live	living	lived	has lived
38	close	closer	closest	the closer

Sharks have always been described as cruel and murderous creatures. Lots of feature films (39) _____ to demonstrate how dangerous they are to people. Though some kinds of sharks really can attack people, the number of people (40) _____ or killed by sharks is very small. Marine biologists insist that most sharks are (41) _____ to people. Along with that, the shark is one of the most ancient creatures of the world, and it should (42) _____ by all countries.

	A	B	C	D
39	shoot	shot	have shot	have been shot
40	injure	injured	injuring	injury
41	harm	harmed	harmless	unharmed
42	protect	is protected	was protected	be protected